

Yusuf Abdulle

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Education

King's College London, PhD - Biostatistics and Health Informatics - *London, UK* **2024 – 2028**

- Thesis: Exploiting the electronic health records and multimodal data to model patterns of onset and progression in Motor Neurone disease (MND/ALS)

University College London, MSc Clinical Neuroscience – Merit - *London, UK* **2020 – 2021**

University of Brighton, BSc (Hons) Biomedical Science – First Class Honours - *Brighton, UK* **2017 – 2020**

Research Experience

Research Assistant | *University College London* **Sep 2023 – Sep 2024**
London, UK

- Analysed AI-induced health disparities across 4 demographic dimensions (race, gender, age, socioeconomic status) and developed quantification tools.
- Worked on Machine Learning research looking at Vision Language Models, benchmarking LLM performance for healthcare, published in venues such as ACL 2024 and IJCAI 2024
- Provided mentorship and research project supervision for undergraduate students and interns.

Translational Data Scientist | *BioCortex* **Oct 2022 – Apr 2023**
London, UK

- Designed first-principle computer simulations to identify clinical trial pitfalls, improving patient response and reducing costs.
- Developed and deployed an automated benchmarking tool to production that predicted DNA sequences against known databases, leading to improved internal prediction results.

Research Assistant | *University College London* **Sep 2021 – Oct 2022**
London, UK

- Architected Graph Neural Networks using PyTorch Geometric and Deep Graph Library to classify MRI-derived adjacency matrices for Frontotemporal Dementia and ALS detection.
- Engineered AWS-based pipeline using EC2 and S3 for mass processing 3T MRI scans, significantly reducing processing time.
- Partnered with external stakeholders (Haleon) to develop mathematical models (Monte Carlo and Markov Chain) to improve patient flow in healthcare pathways.

Journal Articles & Preprints

- **Abdulle, Y.***, Wu, J.*, et al. "Characteristics and Early Diagnosis of Motor Neuron Disease (MND) in 67 million individuals in England" - *medRxiv* (2025).
- Kim, Y., **Abdulle, Y.**, Wu, H. "BioHopR: A Benchmark for Multi-Hop, Multi-Answer Reasoning in Biomedical Domain" - *Findings of ACL* (2025).
- Kim, Y., Wu, J., **Abdulle, Y.**, et al. "Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs" - *TAI4H (Springer LNCS)* (2024).
- Kim, Y., Wu, J., **Abdulle, Y.**, Wu, H. "MedExQA: Medical Question Answering Benchmark with Multiple Explanations" - *BioNLP Workshop* (2024).
- Spears, S. D., **Abdulle, Y. F.**, et al. "Understanding neck collar preferences and user experiences in motor neuron disease" - *Disability and Health Journal* (2024).
- Groves, E., **Abdulle, Y.**, et al. "Benchmarking and Analyzing In-context Learning, Fine-tuning and Supervised Learning for Biomedical Knowledge Curation" - *VLDB Workshops* (2024).
- Budhdeo, S., **Abdulle, Y.**, et al. "Scoping review of knowledge graph applications in biomedical and healthcare sciences" - *medRxiv* (2023).

Conference Presentations & Posters

- **International Symposium for ALS and MND 2025** (San Diego): Characteristics and early diagnosis on MND in 67 million individuals in England: A comparative study on phenotyping models derived by AI, Knowledge Graph and the MND Association Red Flags tool.
- **ACL 2025** (Vienna): BioHopR: A Benchmark for Multi-Hop, Multi-Answer Reasoning
- **HealTAC 2025** (Glasgow): Can GPT-4 be a good red flagger for MND? (Poster/Presentation).

- **HealTAC 2025** (Glasgow): A Large Language Model based Framework for Dementia Related Hypothesis Generation.
- **ENCALS 2025** (Turin): Early ALS phenotyping and COVID-19 Survival Analysis Using EHR Data.
- **MICCAI 2024** (Marrakesh): Enhancing human-computer interaction in chest x-ray analysis using vision and language model with eye gaze patterns.
- **MICCAI 2024** (Marrakesh): Organiser, Foundation Models For Medical Imaging (FOMMIA)
- **TAI4H 2024** (Marrakesh): Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs.
- **BioNLP Workshop 2024** (Bangkok): MedExQA: Medical Question Answering Benchmark with Multiple Explanations.
- **OHBM 2023** (Vancouver): Using a PET neurotransmitter atlas to probe associations in Alzheimer's disease and MCI.
- **EAN Congress 2023** (Budapest): Using a PET atlas to probe neurotransmitter-disease associations in Mild Cognitive Impairment and Alzheimer's disease.

Technical Skills

- **Programming:** Python, R, Bash.
- **Libraries:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, PyTorch Geometric, NetworkX.
- **Technologies:** Git, PostgreSQL, AWS (EC2, S3), GCP.
- **Languages:** English, Somali.

Honors & Awards

- EPSRC Centre for Doctoral Training in Data-Driven Health (DRIVE-Health) Fellowship (2024).
- Santander Summer Studentship (2019).